EverTried

Powering India's Blue-Collar Workforce

Technical Report

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Hackathon Project Submission

SDG-8: Decent Work & Economic Growth

1. Executive Summary

EverTried is a full-stack digital employment ecosystem designed to directly connect India's blue-collar workforce with job opportunities — eliminating intermediaries, reducing exploitation, and enabling fair, fast, and reliable employment at scale.

The platform addresses four systemic failures in the informal labor market:

- Income instability and unpredictability for daily-wage workers
- Exploitation by middlemen and unverified contractors
- Absence of digital identity and formal work history
- Low digital literacy limiting platform access

By combining AI-powered smart matching, voice-based interaction, hyperlocal geo-filtering, and a dual job-routing architecture, EverTried directly advances UN Sustainable Development Goal 8 — Decent Work and Economic Growth.

Project Name	EverTried — No Middlemen, Just Work
Domain	Blue-collar workforce digitization, informal sector employment
SDG Alignment	SDG-8: Decent Work & Economic Growth
Core Stack	React.js · Node.js/Express · MongoDB · Firebase · Fast2SMS/Twilio
Key Innovation	Dual Job Routing System + AI Matching Engine + Voice Search

2. Problem Statement

India's informal labor sector employs over 400 million workers, yet remains largely undigitized and vulnerable to systemic exploitation. The following critical problems are identified:

2.1 Core Challenges

- Unstable and unpredictable daily income with no forecasting mechanisms
- Rampant exploitation by middlemen and contractors taking disproportionate commissions
- Proliferation of fake or unreliable job listings on existing platforms
- No formal work history, resume, or verifiable track record for workers
- Low digital literacy creating significant barriers to app adoption
- Absence of insurance, safety nets, or financial inclusion tools
- Limited career growth pathways for skill development

2.2 Failure of Existing Solutions

Current platforms (e.g., generic job boards) are inadequate because they:

- Function only as passive listing boards with no intelligent matching
- Have poor UX for low-literacy users with no voice or regional language support
- Provide no trust verification or fraud prevention mechanisms
- Offer no financial safety layers or social support infrastructure

3. Proposed Solution & Architecture

3.1 Solution Overview

EverTried is architected as a complete employment ecosystem — not merely a job portal. The core philosophy is: "Direct, trusted, and intelligent job matching for blue-collar workers." The platform integrates an AI matching engine, multi-role access control, a dual job-routing system, and multi-channel notifications to create an end-to-end employment lifecycle.

3.2 High-Level System Architecture

The platform follows a three-tier client-server architecture with an integrated AI layer and external service APIs:

```
Frontend (React.js / React Native)
    ↓  REST API calls (JSON over HTTPS)
Backend API Layer (Node.js / Express.js)
    ├── Auth Service  ├── Job Service  ├── Matching Engine  └── Notification
Service
    ↓  Mongoose ODM
Database Layer (MongoDB Atlas)
    ├── Users  ├── Jobs  └── Applications
    ↓  External Integrations
External Services: SMS API · Firebase FCM · Google Maps · WhatsApp API
```

3.3 Role-Based Architecture

The system implements strict role-based access control (RBAC) across three actor layers:

- Worker Layer: Mobile/web UI for job discovery, application, and notifications
- Employer Layer: Dashboard for job posting, applicant management, and worker rating
- Coordinator Layer: Admin interface for large-scale job decomposition, worker assignment, and progress monitoring

3.4 Dual Job Routing System (Key Innovation)

The platform's primary differentiator is an automatic dual-routing mechanism that classifies jobs at creation time and routes them through the appropriate workflow:

Job Type	Small Scale	Large Scale
Routing	Direct → Available Worker	Employer → Coordinator → Sub-jobs → Workers
Oversight	Employer manages directly	Coordinator monitors all sub-tasks

4. Database Design

4.1 MongoDB Collections

The data layer uses MongoDB Atlas with three primary collections designed for document-oriented flexibility and geospatial querying:

Users Collection

Stores all platform actors with a discriminated role field. Location is stored as a GeoJSON Point to support proximity queries.

```
{   _id: ObjectId,   name: String,   phone: String,           // Primary identifier
(no email required)   role: "worker | employer | coordinator",   skills: [String],
// e.g. ["plumbing", "electrical"]   location: {     type: "Point",     coordinates:
[lng, lat]   },   rating: Number,          // Aggregate score 0-5   availability:
Boolean,   workHistory: [{ jobId, completedAt, rating }] }
```

Jobs Collection

Tracks the full lifecycle of each job from creation to completion. The type field drives the dual routing logic.

```
{   _id: ObjectId,   title: String,   type: "small | large",   postedBy: ObjectId,
// Employer ref   assignedTo: ObjectId,     // Worker ref (or null)   coordinatorId:
ObjectId,  // For large jobs   requiredSkills: [String],   workerCount: Number,
location: { type: "Point", coordinates: [lng, lat] },   status: "open | in-progress |
completed",   createdAt: Date,   completedAt: Date }
```

Applications Collection

Junction collection tracking worker application lifecycle for each job.

```
{   _id: ObjectId,   jobId: ObjectId,   workerId: ObjectId,   status: "applied |
accepted | rejected",   appliedAt: Date }
```

4.2 Indexing Strategy

- 2dsphere index on location fields (Users and Jobs) for efficient geo-queries
- Compound index on (status, type) in Jobs for filtered listing queries
- Index on (jobId, workerId) in Applications for fast lookup
- Index on phone in Users for authentication lookups

5. Feature Specification

The platform delivers a tiered feature set categorized by implementation priority:

Feature	Description	Priority
Multi-Role Authentication	Separate login flows for Workers, Employers, and Coordinators	Core
Job Posting System	Small-scale jobs routed directly to workers; large-scale jobs via coordinators	Core
AI Matching Engine	Skill, location, availability, and rating-based smart matching	Core
Hyperlocal Discovery	Jobs within 1–5 km radius using geo-queries	Core
Notification Engine	SMS (Fast2SMS / Twilio) and push (FCM) alerts on job events	Core
Worker Profile & History	Digital portfolio: skills, ratings, completed job log	Core
Trust & Rating System	Bidirectional rating for workers and employers post-job	Core
Voice-Based Search	Regional-language speech input to remove literacy barrier	Advanced
WhatsApp Integration	Job alerts and one-click apply via WhatsApp	Advanced
Income Stability Engine	ML-driven earnings prediction; job suggestions in low-income periods	Advanced
Dual Job Routing	Automatic routing logic: small → direct worker, large → coordinator sub-jobs	Core

6. REST API Reference

All endpoints are prefixed with /api/v1 and return JSON. Authentication uses JWT Bearer tokens issued on login.

Method	Endpoint	Description
POST	/api/auth/register	Register worker, employer, or coordinator
POST	/api/auth/login	Authenticate and receive JWT token
POST	/api/jobs	Create a new job posting
GET	/api/jobs	Retrieve all open jobs (with filters)
GET	/api/jobs/nearby	Fetch jobs within radius (lat, lng, km)
POST	/api/jobs/:id/apply	Worker applies to a job
POST	/api/jobs/:id/assign	Employer assigns job to a worker
GET	/api/workers/profile	Get authenticated worker's profile
PUT	/api/workers/availability	Update worker availability status
POST	/api/ratings	Submit post-job rating
GET	/api/match/:jobId	Run matching engine for a specific job

7. Matching Engine

7.1 Core Algorithm

The matching engine is triggered after job creation and evaluates available workers against three weighted criteria:

- Skill Match: Worker's skills array must have a non-zero intersection with requiredSkills
- Location Proximity: Worker's GeoJSON coordinates must fall within the job's configured radius (default 5 km); implemented via MongoDB $geoNear aggregation
- Availability: Worker's availability flag must be true

7.2 Scoring Formula (Basic Version)

```
score = (skill_overlap_ratio × 0.4) + (proximity_score × 0.3) + (rating / 5 × 0.3)
```

Workers are ranked by descending score. The top N matches are notified based on workerCount.

7.3 Advanced Extensions

- Rating Weight: Higher-rated workers get boosted score in competitive matches
- Past Job Success Rate: Historical completion rate incorporated as a reliability signal
- Income Stability Engine: ML model predicts worker earnings gaps and proactively surfaces relevant jobs

8. Application Workflows

8.1 Worker Flow

1. Register via simple form or voice input; select skills and set location
2. Browse nearby jobs or receive real-time SMS/push notification on new matches
3. Apply to a job with one click (no resume required)
4. Employer reviews applicants and assigns job
5. Worker completes job and receives payment
6. Both parties rate each other; digital work history is updated

8.2 Employer Flow — Small Scale

7. Post a job with title, required skills, location, and worker count
8. Matching engine notifies nearby available workers automatically
9. Employer reviews matched worker profiles and selects
10. Job status updates to in-progress; employer tracks completion
11. On completion, employer rates worker

8.3 Employer Flow — Large Scale

12. Post a large-scale job; system routes to available coordinator
13. Coordinator decomposes the job into smaller role-specific sub-jobs
14. Sub-jobs are broadcast to workers matching each role's skills
15. Coordinator monitors all sub-job statuses on a unified dashboard
16. On full completion, employer rates coordinator; coordinator rates workers

8.4 Notification Flow

Job Created → Backend Processes Job → Matching Engine Runs → SMS + FCM Push Sent to Top Matches → Worker Opens App / WhatsApp → Worker Applies → Employer Notified → Assignment Confirmed

9. Technology Stack & Deployment

9.1 Stack Summary

Layer	Technology	Justification
Frontend	React.js (Web), React Native (Mobile)	Component reusability; shared logic across platforms
Backend	Node.js + Express.js	Non-blocking I/O for high concurrency job events
Database	MongoDB Atlas	Flexible schema; native GeoJSON for location queries
AI Layer	Custom Node.js scoring + planned ML model	Lightweight matching; extensible to ML pipeline
Auth	JWT + bcrypt	Stateless authentication; password hashing
Notifications	Fast2SMS / Twilio (SMS), Firebase FCM (Push)	Wide reach; no smartphone required for SMS
Maps	Google Maps API	Geo-coding, proximity calculation, demand heatmap
Deployment	Vercel (Frontend), Render/Railway (Backend), MongoDB Atlas (DB)	Low-cost, scalable hosting for hackathon and beyond

10. Competitive Analysis

The table below benchmarks EverTried against existing platforms across the feature dimensions that matter most to the blue-collar workforce:

Feature	Existing Platforms	EverTried
Job Listings	✓	✓
Voice Search	✗	✓
Trust & Rating System	✗	✓
Instant Hiring	✗	✓
Income Prediction	✗	✓
Worker Safety Layer	✗	✓
Digital Work History	✗	✓
WhatsApp Integration	✗	✓
Dual Job Routing	✗	✓
Micro-Finance Support	✗	✓ (Planned)

EverTried's combination of voice search, instant hiring, trust scoring, income prediction, and dual routing constitutes a complete ecosystem — not merely a feature upgrade on existing job boards.

11. Impact Assessment

11.1 Economic Impact

- Reduces worker income instability through predictive job recommendations
- Increases employment cycle speed via instant hiring (minutes vs. days)
- Cuts middlemen commission losses, returning earnings directly to workers
- Enables micro-finance and savings tools linked to verified work history

11.2 Social Impact

- Reduces exploitation and power asymmetry between workers and contractors
- Builds worker dignity through verifiable digital identity and work history
- Voice and WhatsApp integration achieves inclusion for low-literacy users
- Community network enables worker-to-worker referrals and mutual support

11.3 Digital Impact

- Brings an estimated 400M+ informal workers into the digital economy
- Establishes a portable digital resume usable for future financial products
- Platform data enables government labor welfare scheme integration

12. Future Scope

- Government API integration for labor welfare scheme eligibility checks
- AI-based skill certification and course recommendations
- Accident insurance and SOS emergency button with real-time alerts
- Multilingual expansion across 15+ Indian regional languages
- Offline kiosk support via USSD for rural areas with no internet access
- Micro-finance products: daily savings tracking and loan eligibility scoring
- Real-time workforce demand heatmap for city-level labor planning

13. Conclusion

EverTried represents a paradigm shift in how India's blue-collar workforce accesses employment. By eliminating intermediaries, introducing intelligent AI-powered matching, providing multi-channel accessibility (app, SMS, WhatsApp, voice), and implementing a novel dual job-routing architecture, the platform addresses the full spectrum of systemic failures in the informal labor market.

The project directly advances SDG-8 (Decent Work and Economic Growth) and is architected for scale — from a hackathon MVP to a nationwide deployment capable of transforming the livelihoods of millions.

"From survival to stability to growth — EverTried enables it all."